



Real-time lane-level abnormal traffic detection on freeways using sparse telematics data

Shixiao Liang^a, Chengyuan Ma^a,^{*} Pei Li^b, Haotian Shi^{c,a}, Jiayi Liu^a, Hang Zhou^a,
Keke Long^a, Bofeng Cao^d, Todd Szymkowski^e, Xiaowei Shi^f, Xiaopeng Li^a

^a Department of Civil & Environmental Engineering, University of Wisconsin-Madison, Madison, WI, United States

^b Department of Civil & Architectural Engineering and Construction Management, University of Wyoming, Laramie, WY, United States

^c Department of Traffic Engineering & Key Laboratory of Road and Traffic Engineering of Ministry of Education, Tongji University, Shanghai, China

^d Department of Electrical & Computer Engineering, University of Wisconsin-Madison, Madison, WI, United States

^e Wisconsin Department of Transportation, Madison, WI, United States

^f Department of Civil & Environmental Engineering, University of Wisconsin-Milwaukee, Milwaukee, WI, United States

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ABSTRACT

Real-time abnormal traffic detection is critical in intelligent transportation systems because traditional abnormal traffic notifications often suffer delays and lack specific, lane-level location information, which can lead to safety risks and economic losses. This paper proposes a real-time, lane-level abnormal traffic detection approach for freeways that only leverages sparse telematics trajectory data. In the offline stage, the historical trajectories are discretized into spatial cells using vector cross-product techniques, and then are used to estimate a vehicle intention distribution and select an alert threshold by maximizing the F1-score against official crash reports used as proxy labels for abnormal events. In the online stage, incoming real-time telematics records are mapped to these cells and scored for three modules: transition anomalies, speed deviations, and lateral maneuver risks, with scores accumulated into a cell-specific risk map. When any cell's risk exceeds the alert threshold, the system issues a prompt warning. Relying solely on telematics data, this real-time and low-cost solution is empirically evaluated using these report-based labels in Wisconsin, achieving a 75% identification rate with accurate lane-level localization, an overall accuracy of 96%, an F1-score of 0.84, and a false alarm rate of only 0.6% among non-crash cases, while also detecting 13% of crashes more than 3 min before the notification time.

1. Introduction

Real-time abnormal traffic detection plays a critical role in enhancing traffic safety and enabling timely emergency responses (Huang et al., 2020; Hussain et al., 2024; Yuan et al., 2018; Li and Abdel-Aty, 2022a; Li et al., 2020a). Reducing abnormal traffic detection and clearance time by just one minute can yield significant safety and efficiency benefits. According to the Federal Highway Administration (2010), each additional minute of crash duration increases the likelihood of secondary crashes by approximately 2.8%, while reducing clearance time by one minute can cut total traffic delay by up to four minutes. These time savings also carry measurable economic significance: in Hong Kong, crash-related traffic delays in 2021 cost society approximately US \$11.02 million, corresponding to an average of about US \$0.26 per vehicle-minute of lost productivity and fuel

consumption (Lian and Loo, 2024). Thus, even a one-minute improvement in abnormal traffic detection and response could prevent multiple secondary crashes and save substantial societal costs, underscoring the critical importance of rapid, accurate alerts.

Abnormal traffic detection refers to the task of identifying when and where abnormal traffic occurs based on available traffic data (e.g., aggregated traffic measures or vehicle trajectories). In this paper, “abnormal traffic” denotes safety-relevant, nonrecurrent disturbances in freeway operations that produce measurable deviations in lane-level behaviors, including but not limited to crashes, lane-blocking incidents, disabled vehicles, and other unexpected events. Crashes are operationally the most important and most frequently documented subset of these disturbances, but they do not define the methodological task itself. Current abnormal traffic detection approaches fall into two broad

* Corresponding author.

E-mail addresses: sliang85@wisc.edu (S. Liang), cma97@wisc.edu (C. Ma), pei.li@uwyo.edu (P. Li), shihaotian95@tongji.edu.cn (H. Shi), jliu2487@wisc.edu (J. Liu), hzhou364@wisc.edu (H. Zhou), klong23@wisc.edu (K. Long), bcao44@wisc.edu (B. Cao), todd.szymkowski@dot.wi.gov (T. Szymkowski), tomshi@uwm.edu (X. Shi), xli2485@wisc.edu (X. Li).

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categories based on the criteria of detection (ElSahly and Abdelfatah, 2022). The first category focuses on using macroscopic, aggregated traffic indicators such as flow, density, and speed, to identify abnormal traffic conditions (Zhu et al., 2021; Zhang et al., 2023; Querfurth et al., 2025). While effective at identifying major crashes that cause network-wide congestion, these techniques often suffer from substantial detection delays because they rely on traffic disturbances gradually propagating through the network. Their spatial resolution is inherently coarse, typically limited to corridor-level anomalies rather than precise lane-level roadway segments, which highlights a persistent gap in fine-grained localization research. Existing macroscopic studies rarely evaluate the exact spatial accuracy of localization, leaving the problem underexplored. Nonetheless, many operational and safety applications require much finer spatial resolution than macroscopic indicators can provide, because effective countermeasures rely on knowing the exact lane affected by an abnormal traffic event. Achieving lane-level precision is therefore critical for enabling proactive lane changes, maintaining smoother traffic flow, and reducing the likelihood of secondary crashes.

On the other hand, microscopic methods employ either roadside sensing or vehicle trajectory data to detect abnormal maneuvers at the lane level (Rocky et al., 2024). Approaches relying on roadside cameras, LiDAR, or other fixed sensors can provide highly accurate localization but require costly infrastructure and dense deployments, which limits scalability (Qu et al., 2024b; Ali et al., 2023). Methods based on high-fidelity vehicle trajectories achieve comparable precision but often depend on proprietary or high-frequency data that are difficult to obtain in real time. Moreover, even when high-quality vehicle trajectory data are available, few studies have achieved lane-level accuracy without relying on additional sensor fusion. Therefore, these limitations highlight the need for a scalable and cost-effective data source that can capture vehicle-level dynamics while avoiding the deployment burden of dense roadside sensors or proprietary high-frequency datasets.

Telematics data, collected from in-vehicle IMU, GPS, and communication systems, can provide time-stamped trajectories and speed profiles of individual vehicles, enabling microscopic observation of traffic dynamics at a large scale (Herfort et al., 2023; Kan et al., 2019; Zhang and Abdel-Aty, 2022). Recent industry reports further indicate that telematics systems have already reached an estimated 79 percent penetration among new vehicles worldwide in 2024, suggesting that such data sources are becoming increasingly ubiquitous (Cederqvist, 2025). Owing to their wide availability and low deployment cost, such data support low-latency, real-time abnormal traffic detection at lane-level granularity while remaining scalable to nationwide deployment. However, although modern GPS technology can achieve sub-meter positioning accuracy, the trajectories collected through large-scale telematics platforms are typically temporally sparse because most providers transmit data at intervals of three to five seconds. On freeways, where vehicles travel at high speeds, this upload frequency results in long spatial gaps between consecutive points. In addition, only about five to seven percent of vehicles in the total traffic stream are represented in these data, which further limits the completeness of the reconstructed traffic state (Cederqvist, 2025). Leveraging such incomplete trajectories to identify abnormal traffic patterns remains challenging. To improve spatial precision, scholars have begun developing machine learning frameworks that snap sparse vehicle telematics records to detailed lane geometries for microscopic risk inference (Jin et al., 2023; Li and Abdel-Aty, 2022b; Kubin et al., 2021; Dai et al., 2023; Qu et al., 2024a; Santos et al., 2022; Huang et al., 2020; Li et al., 2020b; Li and Abdel-Aty, 2022a). While these learning-based approaches achieve promising results, they often lack physical interpretability. Recent studies have explored physics-enhanced learning frameworks that combine data-driven models with physical priors (Lee and Chang, 2025; Liang et al., 2025), yet their implementation in abnormal traffic detection still requires high-frequency data or specialized sensors (e.g., built-in accelerometers (Sharma et al., 2016)). Despite these advances, a gap

remains for lightweight and scalable methods that can operate effectively on sparse telematics data. Furthermore, most machine learning approaches rely on large labeled datasets and numerous contextual features (e.g., speed limits, weather, or infrastructure conditions), which constrains their generalizability and precision when applied solely to telematics data.

Recent studies on automatic incident detection (AID) have explored a broad range of data sources and modeling paradigms. Trajectory-based machine-learning approaches have been used for real-time highway abnormal traffic detection from vehicle movement data (Roy and Hasan, 2026), while spatiotemporal traffic-feature analysis has also been shown to reduce automatic accident detection time (Morianio et al., 2024). Related studies have further examined real-time freeway crash detection using connected vehicle waypoint data (Kandiboina et al., 2025), freeway abnormal traffic detection using dual-state traffic factor representations (Zhang et al., 2025), and more recent explainable AID frameworks based on spatiotemporal self-supervised learning and multi-probing classifiers (Zhai et al., 2025). These studies demonstrate the growing interest in data-driven AID, but they also continue to rely on richer trajectory resolution, stronger supervision, or more complex model structures than the sparse telematics setting considered in this study. Consequently, detecting abnormal traffic from sparse and low-granularity telematics trajectories remains an open challenge (Pang et al., 2013; Roy and Hasan, 2024; Kandiboina et al., 2025). To make effective use of these trajectories, one must extract and accumulate subtle abnormal driving behaviors such as sudden lane changes and decelerations that are often observed upstream of the abnormal traffic site as drivers maneuver to bypass the blockage (Shi et al., 2025). To address this, we propose a novel model-free framework that separates the offline model-building stage from the real-time detection stage to enable an efficient and adaptive abnormal traffic detection process. In the offline stage, raw telematics records are matched to lane geometries and discretized into spatial cells to mitigate sampling sparsity. Historical trajectories are used to construct a vehicle intention distribution under normal conditions. The alert threshold is then determined by maximizing the F1-score, with official crash reports serving as proxy labels and an external evaluation benchmark rather than as the definition of the task (Powers, 2020). After data preparation, in the online detection stage, each incoming telematics record is assigned to its corresponding cell based on the same algorithm in the offline stage. For each assigned cell, we compute three risk metrics. The first metric measures how unexpected the observed transition is relative to the vehicle intention distribution. The second metric quantifies the degree of speed deviation from normal traffic conditions. The third identifies abnormal lane change maneuvers. These risk metrics are weighted summed to update a dynamic risk map every second. Whenever the accumulated risk in any cell exceeds the safety alarm threshold, the system issues a prompt warning with precise lane-level localization.

The main contributions of this work are summarized as follows:

1. We propose a real-time abnormal traffic detection framework that directly uses sparse telematics data, addressing the practical challenge of low-frequency and low-penetration observations.
2. To enable robust online detection under data sparsity, we develop a behavior-based modeling and risk-fusion scheme that integrates intention-distribution learning with abnormality accumulation.
3. When evaluated against official crash-report proxy labels, our method achieves a 75% identification rate with only 0.6% false positives among non-crash cases, while also providing accurate lane-level localization and early warning capability.

The remainder of this paper is organized as follows. Section 2 details our methodology. Section 3 describes the dataset and experimental setup and presents the results. Section 4 provides additional sensitivity and supplementary statistical analyses. Section 5 discusses the findings. Section 6 concludes and outlines future research directions.

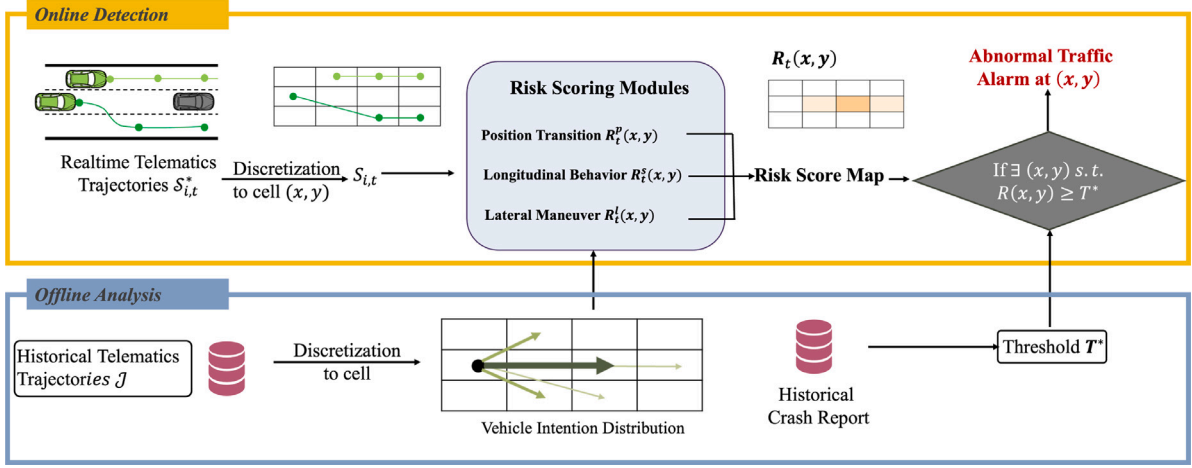


Fig. 1. Overview of the real-time abnormal traffic detection framework.

2. Methodology

In this section, we address the problem of detecting freeway abnormal traffic in real time using sparse telematics data collected from CVs. The task relies on two types of information: (i) large-scale historical CV trajectories and official crash reports, which are used to learn normal driving patterns and to calibrate model parameters, and (ii) streaming real-time telematics data, which provide instantaneous observations of vehicle behavior. Our objective is to design a lane-level abnormal traffic detection method to continuously monitor freeway segments and identify abnormal driving behaviors. Official crash reports are used only to calibrate and evaluate the framework because they provide the most systematic archival records of safety-relevant abnormal events in our study area. The final output of the methodology is a real-time abnormal traffic detection signal that specifies whether an abnormal traffic has occurred and, if so, the corresponding lane-level location.

2.1. Framework overview

Fig. 1 illustrates the overall structure of the proposed abnormal traffic detection framework, which consists of an offline analysis stage and an online detection stage. The offline stage relies on a large-scale historical telematics dataset $\mathcal{J} = \{S_{i,t}^* \mid i \in \mathcal{I}, t = 1, \dots, T\}$, where $\mathcal{I}$ is the index set of all vehicles and T is the total number of time steps in the dataset. Each raw telematics record is given by $S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t})$, denoting the reported latitude, longitude, and instantaneous speed of vehicle i at time t . These records are transformed into discretized states $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$, where $x_{i,t}$ denotes the lane index and $y_{i,t}$ denotes the longitudinal segment index on the roadway based on the map geometry of the study corridor. The set of all possible cell locations is denoted by $\mathcal{S} = \{(x, y)\}$, so that the spatial component of each discretized state satisfies $(x_{i,t}, y_{i,t}) \in \mathcal{S}$. Using the historical dataset $\mathcal{J}$, the offline stage learns the vehicle intention distribution that characterizes normal longitudinal and lateral driving behaviors under non-crash conditions. In addition, official crash reports aligned with telematics trajectories are used to calibrate the alert threshold T^* and other model parameters required for real-time detection.

During online operation, streaming telematics records $S_{i,t}^*$ arrive sequentially from a time-varying set of vehicles $i \in \mathcal{I}_t$. Each record is converted into a discretized state $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$. For each time step, the system evaluates three instantaneous risk components at the visited cell $(x_{i,t}, y_{i,t})$: the position transition risk $R_{i,t}^p$, the speed deviation risk $R_{i,t}^s$, and the lateral maneuver risk $R_{i,t}^e$. These per-vehicle risks are aggregated over all vehicles in the same cell and accumulated through time to form a dynamic risk map $R_t(x, y)$. Whenever a vehicle traverses a cell, the accumulated risk at that cell is reset, indicating

that the region has been directly observed and verified to be safe. An abnormal traffic alarm is issued when the accumulated risk at any cell exceeds the offline-calibrated threshold T^* , and the corresponding cell (x, y) is reported as the detected location. Through this integration of historical intention patterns and real-time behavioral evidence, the framework achieves lane-level abnormal traffic detection under sparse connected-vehicle penetration.

2.2. Offline calibration stage

2.2.1. Discretization

Discretization maps each raw telematics record $S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t})$ into a unique cell-level state $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$, where $x_{i,t}$ is the lane index and $y_{i,t}$ is the longitudinal segment index. This mapping replaces free-floating GPS coordinates with a structured grid, reduces raw data complexity, supports efficient computation of transition probabilities, and provides the computational foundation for rapid online risk-score updates. Operating on a fixed grid further enables accumulated risk values to be visualized and updated with minimal overhead even under sparse CV penetration rate.

To determine the lane index $x_{i,t}$, we employ a lane-matching algorithm that assigns each telematics record to its corresponding lane using geometric rules based on publicly available map data such as OpenStreetMap (OSM). Traditional lane-detection techniques such as hidden Markov models, least-squares optimization, or fuzzy logic (Yang et al., 2018; Hansson et al., 2020) are computationally expensive, and image-based approaches (Guo et al., 2024; Rabe et al., 2016) require continuous video streams unavailable in sparse CV data. Given improving GPS accuracy, a geometric method based on vector cross-product provides an efficient and robust alternative.

To perform lane matching, we first construct the reference geometry required for mapping each telematics record to its corresponding lane. Starting from the map-based centerline of the roadway, we obtain them from publicly available digital maps such as OSM. Based on these centerline points, we generate the reference line used for lane determination. For divided or one-way facilities, the centerline directly represents the travel direction; for bidirectional roadways, the centerline is laterally offset by half of the total carriageway width to create separate reference lines for each travel direction. Each reference line is then interpolated into a dense sequence of vertices to ensure consistent spatial resolution for projecting telematics points and computing lateral offsets.

With the reference geometry defined, each telematics record can be assigned to its lane using a geometric lane-matching procedure. For each record $S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t})$, we represent its position $(\text{lat}_{i,t}, \text{lon}_{i,t})$ in the local roadway frame as point A . A perpendicular is dropped from

A to the selected reference line to obtain the foot point C. We then take the next telematics sample of the same vehicle in time $S_{i,t+1}^*$, denoting $(\text{lat}_{i,t+1}, \text{lon}_{i,t+1})$ as point B, and form the vectors $\vec{CA}$ and $\vec{AB}$. Their cross product determines the direction of A relative to the reference line: the sign of its z-component, following the right-hand rule, indicates whether A lies to the left or right of the reference line.

The signed lateral offset is then computed as

$$\delta = \text{signedDist}(A, C),$$

representing the perpendicular distance from A to the reference line, with negative values on the left and positive values on the right. This continuous offset is discretized into the lane index $x_{i,t}$ using

$$x_{i,t} = \left\lfloor \frac{\delta + \frac{W^*}{2}}{w} \right\rfloor + 1,$$

where W^* is the carriageway width associated with the selected reference line and w is the nominal lane width. Adding $W^*/2$ shifts δ from $[-W^*/2, W^*/2]$ to $[0, W^*]$, and dividing by w partitions the cross section into uniformly spaced lanes, with the floor operator producing a one-based integer lane index. The complete procedure is summarized in Algorithm 1.

Algorithm 1 Lane Matching

Require: road segment of interest $(\text{lat}_c, \text{lon}_c)$, road identifier ID_c , telematics records S^* , lane width w , lane count n , search radius r , OSM tags: oneway, dual_carriageway

- 1: filter S' by proximity to $(\text{lat}_c, \text{lon}_c)$ within radius r
- 2: retrieve from OSM the centerline C and tags oneway, dual_carriageway
- 3: compute total width $W \leftarrow w \times n$
- 4: **if** oneway = yes or dual_carriageway = yes **then**
- 5: $C^* \leftarrow C$, $W^* \leftarrow W$, $n^* \leftarrow n$
- 6: **else**
- 7: $d \leftarrow W/4$
- 8: $C_L \leftarrow \text{offset}(C, +d)$, $C_R \leftarrow \text{offset}(C, -d)$
- 9: $C^* \leftarrow \arg \min_{X \in \{C_L, C_R\}} \text{dist}(X, (\text{lat}_c, \text{lon}_c))$
- 10: $W^* \leftarrow W/2$, $n^* \leftarrow n/2$
- 11: **end if**
- 12: **for all** $A = S_{i,t}^* \in S'$ **do**
- 13: $C_A \leftarrow \arg \min_{c \in C^*} \text{dist}(A, c)$
- 14: $B \leftarrow \text{next}(S_{i,t}^*)$ $\triangleright$ next telematics record of same vehicle i
- 15: $\vec{CA} \leftarrow A - C_A$
- 16: $\vec{AB} \leftarrow B - A$
- 17: $z \leftarrow (\vec{CA} \times \vec{AB})_z$
- 18: **if** $z > 0$ **then** $s \leftarrow \text{right}$
- 19: **else** $s \leftarrow \text{left}$
- 20: **end if**
- 21: $\delta \leftarrow \text{signedDist}(A, C_A)$
- 22: $\text{laneIdx} \leftarrow \lfloor (\delta + W^*/2)/w \rfloor + 1$
- 23: record $(\text{laneIdx}, s)$ for A
- 24: **end for**
- 25: **Output:** laneIdx and side s for all telematics records in S'

After assigning each telematics record a lane index $x_{i,t}$, we segment each lane longitudinally into fixed-length intervals, and each telematics record is assigned a longitudinal segment index $y_{i,t}$. The pair $(x_{i,t}, y_{i,t})$ therefore identifies the unique cell visited by vehicle i at time t , completing the cell-level representation $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$. In our dataset, GPS timestamps arrive approximately every 3 s. At typical freeway speeds (approximately 75 mph or 33.5 m/s), a vehicle travels about 100 m within this interval, so we adopt 10 m longitudinal segments when defining the index $y_{i,t}$. In practice, dividing each lane into roughly ten 10 m cells provides sufficient spatial resolution without overcrowding consecutive samples in the same cell. Curvilinear sections

of roadway are not problematic, as OSM provides dense centerline vertices that allow the curved alignment to be well approximated by multiple short polylines. Considering the inherent GPS noise, the resulting longitudinal segmentation remains robust under this geometric interpolation. This speed-adaptive rule yields uniformly occupied cells along each trajectory. By combining the lane index $x_{i,t}$ with the longitudinal segment index $y_{i,t}$, we obtain the final discretized state $S_{i,t}$ with minimal sampling bias.

For illustration, Fig. 2 presents the workflow that transforms raw telematics records into discretized lane-level states. Fig. 2(a) shows the raw telematics data collected from all vehicles. Fig. 2(b) shows the records extracted within the roadway segment of interest. Fig. 2(c) illustrates the lane-matching step, where each raw record $S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t})$ is assigned a lane index $x_{i,t}$. Fig. 2(d) further applies longitudinal segmentation to assign a segment index $y_{i,t}$, yielding the final discretized cell-level state $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$. This example visually summarizes the preprocessing steps used to convert sparse telematics observations into structured lane-level cell states for both the offline and online stages.

2.2.2. Vehicle intention distribution extraction

Through the discretization procedure described above, each raw telematics record $S_{i,t}^*$ is mapped to a cell-level state $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$, providing a consistent spatiotemporal representation for modeling normal driving behavior. This structure enables the extraction of a vehicle intention distribution, which captures the most likely next cell a vehicle will occupy under routine traffic conditions.

For each historical trajectory in $\mathcal{J}$ and for every time step t , we record the transition from the current cell $(x_{i,t}, y_{i,t})$ to the next observed cell $(x_{i,t+1}, y_{i,t+1})$ one timestamp later (3 s). For a fixed origin cell (x, y) , all transitions of the form $(x, y) \rightarrow (x', y')$ are aggregated across all vehicles and all days in the 7-day historical window. Normalizing these aggregated counts yields an empirical transition probability:

$$P((x, y) \rightarrow (x', y')),$$

which represents the estimated statistical expectation that a vehicle in cell (x, y) will move to cell (x', y') in the next 3 s interval.

The resulting vehicle intention distribution characterizes routine longitudinal and lateral movements across the corridor and forms the statistical baseline for our risk analysis, following a goal-prediction framework (Gan et al., 2025). Transitions with high probability encode typical behavior, whereas any observed transition $S_{i,t} \rightarrow S_{i,t+1}$ that deviates substantially from these high-probability patterns is flagged as anomalous and treated as a potential risk factor.

2.2.3. Threshold determination

In the offline stage, in addition to extracting the vehicle intention distribution from historical trajectories, we also use historical crash reports to calibrate the alert threshold for the real-time detection system. For each crash sample in the offline dataset, we run the complete risk-scoring procedure. This produces an accumulated risk map $R_t(x, y)$ for every cell (x, y) over time. For each event, we record the maximum accumulated risk value attained within the affected spatiotemporal region. We then evaluate a range of candidate thresholds T by comparing whether $R_t(x, y) \geq T$ would correctly identify the abnormal traffic in a validation set. For each candidate threshold, we compute precision, recall, and the corresponding F1-score based on the binary detection outcomes. The optimal threshold T^* is chosen as the value that maximizes the F1-score across all validation samples.

During real-time operation, the system continuously maintains the accumulated risk score $R_t(x, y)$ for every cell. An alert is triggered whenever any cell exceeds the offline-calibrated threshold T^* , providing a lane-level early-warning mechanism based on deviations from routine driving behavior. In this study, the alert threshold T^* is calibrated using the summer dataset only and then transferred unchanged to the winter dataset for evaluation.

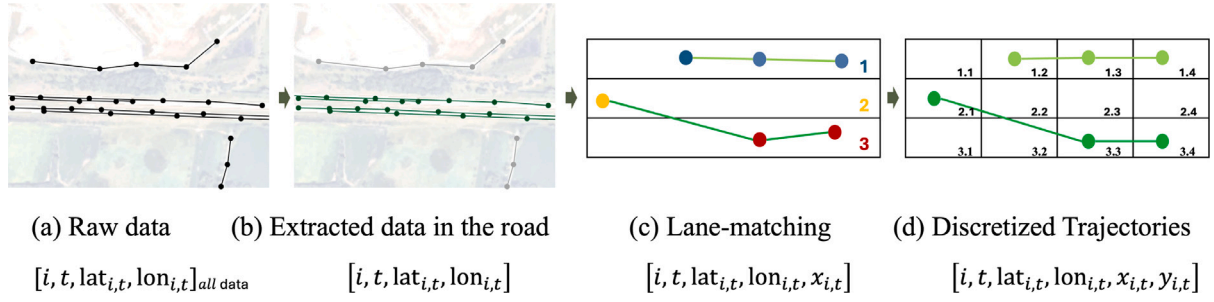


Fig. 2. Illustration of the discretization workflow from continuous vehicle motion to lane-level cell states.

2.3. Online detection stage

The offline calibration stage provides two key components for real-time detection: the vehicle intention distribution $P((x, y) \rightarrow (x', y'))$, which represents normal driving behavior across the discretized grid S , and the optimized alert threshold T^* . These components are then applied to incoming telematics data to dynamically compute a risk score. The risk scoring process translates observed trajectories $S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t})$ into discretized states $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$ and evaluates deviations from normal driving patterns. At each time step t , a vehicle's state at cell $(x_{i,t}, y_{i,t})$ is evaluated by three complementary modules: transition anomaly detection, speed deviation analysis, and lateral maneuver risk assessment. Their weighted sum produces a comprehensive risk score capable of triggering real-time alerts.

These module outputs are continuously accumulated on a spatial risk map $\{A_t(x, y) : (x, y) \in S\}$. To prevent these accumulated values from growing indefinitely, a reset indicator $\phi_t(x, y)$ is applied. When $\phi_t(x, y) = 1$, the cell's accumulated risk $A_t(x, y)$ is cleared, indicating that a new trajectory has passed through and the area is free of immediate hazards. This mechanism maintains numerical stability and ensures that the risk map reflects only recent, unrecovered anomalies. The following subsections describe each module in detail.

2.3.1. Position transition risk

To detect unlikely trajectory changes, we compare each cell transition of a vehicle from $S_{i,t}$ to $S_{i,t+1}$ against the vehicle intention distribution from offline analysis stage. Under normal conditions, vehicles follow high-probability transitions, whereas low-frequency transitions indicate abnormal movement. Therefore, transitions with low empirical frequency signal abnormal behavior. We measure this surprise via a negative log-likelihood transform because the negative log term penalizes the violation of highly probable transitions: when a transition with large expected probability P fails to occur, the term $-\ln(1 - P)$ produces a high risk value. :

$$R_{i,t}^p(x_{i,t}, y_{i,t}) = -\ln(1 - P((x_{i,t}, y_{i,t}) \rightarrow (x_{i,t+1}, y_{i,t+1}))) \quad (1)$$

Here, $P((x, y) \rightarrow (x', y'))$ is the empirical intention distribution estimated from historical trajectories. A high transition probability yields small risk, while rare transitions produce large penalties. A cutoff ϵ_p is applied to exclude transitions whose empirical probability is negligibly small. Such low probabilities may arise from data sparsity or estimation noise in the historical trajectories and typically correspond to infrequently visited cells. Without filtering, these cells can accumulate small but persistent risk values over time during normal driving. Applying this cutoff, together with the periodic cell-level reset described earlier, prevents these residual risks from inflating the overall risk score and ensures numerical stability.

2.3.2. Speed deviation risk

Freeway abnormal traffic often involve sudden speed drops. For each discretized state $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$, we flag scenarios where the instantaneous speed $v_{i,t}$ falls below a reference speed v_{th} :

$$R_{i,t}^s(x_{i,t}, y_{i,t}) = \begin{cases} \frac{v_{th} - v_{i,t}}{v_{th}}, & v_{i,t} \leq v_{th}, \\ 0, & v_{i,t} > v_{th}. \end{cases} \quad (2)$$

2.3.3. Lateral maneuver risk

Frequent lane changes within a confined area can signal lane blockage caused by a crash or hazard. When a lane is temporarily occupied, vehicles often move laterally to adjacent lanes to bypass the obstruction and then return once the lane becomes clear. Hence, consecutive lane-change events detected at the same location over a short period indicate potential crash-related disturbances. We detect a lane-change event by comparing successive lane indices $x_{i,t} \neq x_{i,t-1}$ and assign a risk score for each occurrence:

$$R_{i,t}^l(x_{i,t}, y_{i,t}) = \mathbf{1}\{x_{i,t} \neq x_{i,t-1}\}. \quad (3)$$

The indicator $\mathbf{1}\{\cdot\}$ equals one when a vehicle moves laterally between lanes in one timestep. Cells where lane-change events occur repeatedly over consecutive timesteps accumulate higher risk, reflecting the likelihood of temporary blockage or crash-induced detours in that segment.

2.3.4. Total risk score

The risk components defined above represent vehicle-level quantities evaluated at each time step for the discretized state $S_{i,t}$. To obtain a spatially aggregated measure over the discretized grid, each vehicle's instantaneous risk is mapped to its corresponding spatial cell $(x_{i,t}, y_{i,t}) \in S$ and summed with those of all vehicles currently occupying that same cell. This aggregation produces a cell-level representation of abnormality that reflects collective traffic behavior:

$$\bar{R}_t^m(x, y) = \sum_{i: x_{i,t}=x, y_{i,t}=y} R_{i,t}^m(x_{i,t}, y_{i,t}), \quad m \in \{p, s, \ell\}. \quad (4)$$

Here, m indexes the three modules: position transition risk, speed deviation risk, and lateral maneuver risk. Cells with many vehicles exhibiting similar anomalies naturally accumulate larger cell-level risks, providing robustness against noise in individual trajectories. These aggregated cell-level risks are then combined to form the total instantaneous risk for each cell:

$$\bar{R}_t(x, y) = w_p \bar{R}_t^p(x, y) + w_s \bar{R}_t^s(x, y) + w_\ell \bar{R}_t^\ell(x, y), \quad (5)$$

where the nonnegative coefficients (w_p, w_s, w_ℓ) are calibrated offline.

To capture abnormal behavior that persists over time, we define the accumulated risk for each cell (x, y) by $A_t(x, y)$, initialized with $A_0(x, y) = 0$. The accumulated risk grows as long as anomalies continue to occur near that location. However, when a new trajectory passes through a cell, that cell is directly observed to be navigable, and thus

temporarily hazard-free. To reflect this, a cell-level reset mechanism is introduced:

$$A_t(x, y) = (1 - \phi_t(x, y)) [A_{t-1}(x, y) + \bar{R}_t(x, y)], \quad (6)$$

where $\phi_t(x, y) \in \{0, 1\}$ is a reset indicator equal to 1 iff there exists some vehicle i such that $(x_{i,t}, y_{i,t}) = (x, y)$ at time t . When $\phi_t(x, y) = 1$, the cell risk is cleared to zero because the region has been directly traversed and verified as safe. This prevents false escalation of long-term accumulated risk and maintains numerical stability.

A lane-level abnormal traffic alert is issued the first time

$$A_t(x, y) \geq T^*, \quad (7)$$

where T^* is the threshold selected during offline calibration using historical crash reports. This additive and accumulative structure enables the system to fuse short-term anomalies and sustained abnormal behavior into a coherent spatiotemporal risk metric. Whenever the accumulated risk in any cell exceeds T^* , the system immediately emits an abnormal traffic alert and reports the cell (x, y) as the detected location. We summarize the whole methodology into the following Algorithm 2.

3. Experiment

3.1. Dataset and experimental setup

The dataset used in this study spans the entire United States and covers two meteorologically distinct, continuous one-week periods: a summer week (August 4–10, 2024) and a winter week (January 9–15, 2024). The nationwide summer dataset alone contains approximately 11.4 terabytes of telematics data stored across 208,840 Parquet files, totaling about 1.1×10^{11} GPS observations sampled at 3-second intervals. To complement this large-scale data, the winter dataset adds an additional 215 GB of high-resolution telematics records covering the entire state of Wisconsin. This combination of nationwide coverage, fine-grained temporal sampling, and cross-seasonal variation provides a robust basis for evaluating real-time abnormal traffic detection. The data span a wide range of roadway geometries, regional driving behaviors and traffic densities, allowing rigorous assessment of the system's generalization and operational robustness.

For this study, we restrict our analysis to vehicle trajectories located within Wisconsin, as all crash reports originate from this region. To avoid confounding delays caused by signalized intersections, we further restrict our attention on freeway segments. Fig. 3 illustrates a representative crash case in Milwaukee during August 2024. The left figure shows the spatial location of the crash along the I-94 freeway segment. The right figure presents the corresponding time-space trajectories of vehicles approaching the crash site, with the crash occurring at 06:43 on August 5, 2024. Vehicle trajectories are extracted within a fixed radius (approximately 200 m) centered at the crash point, covering a 25-minute window before and after the abnormal traffic. Each trajectory is color-coded by instantaneous speed. The evident slowdown and clustering of trajectories downstream clearly indicate the congestion induced by the crash.

Based on the filtered trajectories obtained around the crash point, we further construct the vehicle intention distribution for this segment. The telematics data are exceptionally rich. Over the seven-day window, more than 360,000 trajectory points were recorded, with the majority concentrated in the main travel lanes (see Fig. 4(d)). Fig. 4(a) presents the daily distribution of GPS samples, and Fig. 4(b) shows the hourly distribution on the crash day (August 5th). In addition, Fig. 4(c) visualizes the statewide spatial density map of telematics points, highlighting high-coverage regions along major freeway corridors. Together, these distributions demonstrate the dataset's comprehensive temporal and spatial coverage necessary for robust abnormal traffic detection.

Algorithm 2 Abnormal Traffic Detection: Offline and Online Stages

Require: historical telematics dataset $\mathcal{J} = \{S_{i,t}^*\}$, historical crash records, lane width w , lane count n , transition cutoff ϵ_p , speed baseline $v_{th}(x, y)$, module weights (w_p, w_s, w_ℓ) .

Offline Stage

- 1: Discretize all records $S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t})$ into $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$ via map-matching and longitudinal segmentation.
- 2: Estimate empirical transition probabilities $P((x, y) \rightarrow (x', y'))$ from non-crash portions of $\mathcal{J}$.
- 3: Estimate normal speed baselines $v_{th}(x, y)$ for each cell.
- 4: Determine alert threshold T^* by maximizing validation F1-score based on official crash reports.
- 5: Calibrate risk module weights (w_p, w_s, w_ℓ) .

Online Stage

- 6: Initialize accumulated risk $A_0(x, y) \leftarrow 0$ for all $(x, y) \in \mathcal{C}$.
- 7: **for** each time step $t = 1, 2, \dots$ **do**
- 8: Receive telematics records $\{S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t}) : i \in \mathcal{I}_t\}$.
- 9: Discretize each record to $S_{i,t} = (x_{i,t}, y_{i,t}, v_{i,t})$ and retrieve $(x_{i,t-1}, y_{i,t-1})$ if available.
 (Per-vehicle instantaneous risks at $(x_{i,t}, y_{i,t})$)
- 10: **for** each vehicle $i \in \mathcal{I}_t$ **do**
- 11: **Position-transition risk:**

$$R_{i,t}^p(x_{i,t}, y_{i,t}) = \begin{cases} -\ln(1 - P((x_{i,t-1}, y_{i,t-1}) \rightarrow (x_{i,t}, y_{i,t}))) & \text{if } P > \epsilon_p, \\ 0, & \text{otherwise.} \end{cases}$$
- 12: **Speed risk:**

$$R_{i,t}^s(x_{i,t}, y_{i,t}) = \max\left(0, \frac{v_{th}(x_{i,t}, y_{i,t}) - v_{i,t}}{v_{th}(x_{i,t}, y_{i,t})}\right).$$
- 13: **Lateral maneuver risk:**

$$R_{i,t}^\ell(x_{i,t}, y_{i,t}) = \mathbf{1}\{x_{i,t} \neq x_{i,t-1}\}.$$
- 14: **end for**
 (Aggregate per-vehicle risks to cell level)
- 15: **for** each occupied cell (x, y) at time t **do**
- 16: $\bar{R}_t^m(x, y) \leftarrow \sum_{i: x_{i,t}=x, y_{i,t}=y} R_{i,t}^m(x, y) \quad \text{for } m \in \{p, s, \ell\}$
- 17: $\bar{R}_t(x, y) \leftarrow w_p \bar{R}_t^p(x, y) + w_s \bar{R}_t^s(x, y) + w_\ell \bar{R}_t^\ell(x, y)$
- 18: **end for**
 (Update accumulated risk map and detect abnormal traffic)
- 19: **for** each cell $(x, y) \in \mathcal{C}$ **do**
- 20: $\phi_t(x, y) \leftarrow \mathbf{1}\{\exists i : x_{i,t} = x, y_{i,t} = y\} \quad \triangleright$ reset if any vehicle enters (x, y)
- 21: $A_t(x, y) \leftarrow (1 - \phi_t(x, y)) [A_{t-1}(x, y) + \bar{R}_t(x, y)]$
- 22: **if** $A_t(x, y) \geq T^*$ **then**
- 23: EMITCRASHALERT $((x, y), t)$
- 24: **end if**
- 25: **end for**
- 26: **end for**

3.2. Discretization and vehicle intention distribution extraction

In the offline stage, following Section 2.2 and Algorithm 1, we first discretize the area surrounding the crash location as shown in Fig. 5(a) and (b). In the spatial dimension we discretize laterally by lane and longitudinally using speed-dependent segments. For lateral segmentation, if OSM provides explicit lane-width tags, we use those values; otherwise we default to the U.S. freeway standard lane width of 3.7 m (Hancock and Wright, 2013). For longitudinal segmentation in this example, with a posted speed limit of 75 mph (≈ 33.5 m/s),

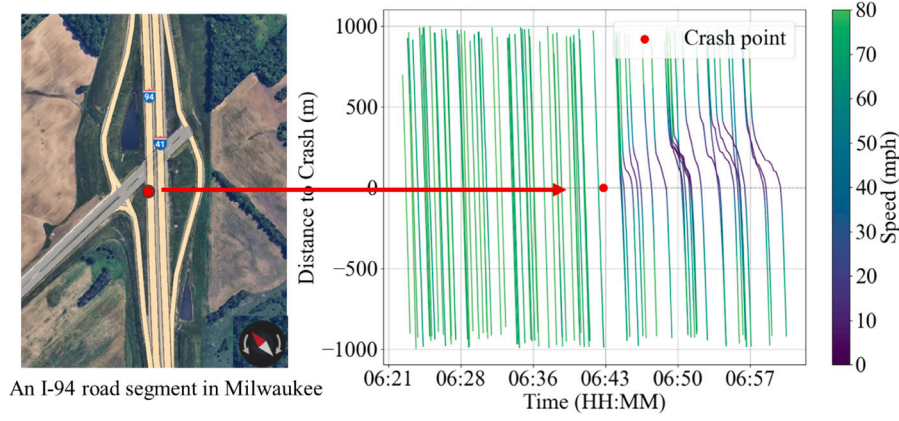


Fig. 3. Crash location and corresponding time-space diagram.

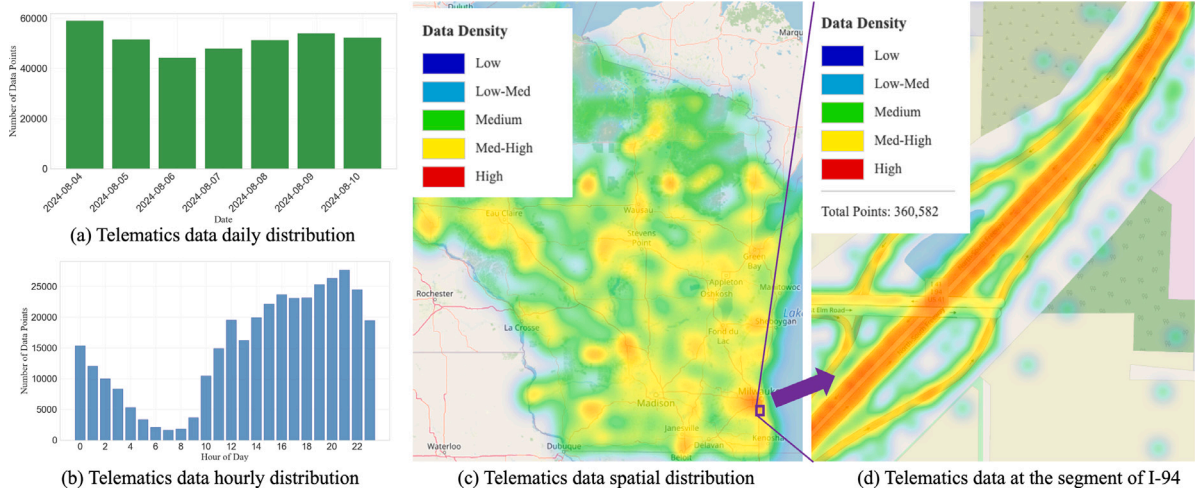


Fig. 4. Telematics data coverage and density for the freeway segment.

we use 10 m cells so that a vehicle travels roughly 10 cells between three-second samples. This rule-based design balances spatial resolution against the sparsity of three-second telemetry. Considering the noise of telematics data, moderate variations in lane width or segment length have minimal effect on detection performance.

Next, we apply this process to the 360,000+ trajectory points collected over the seven-day period. After filtering by heading north degree from telematics data and snapping to the derived road centerline based on OSM road geometry, we isolate the crash-affected segment freeway. Using the vehicle intention distribution extraction method in Section 2.2, we count transitions from each source cell and normalize them to derive the vehicle intention distribution. Fig. 5(c) highlights one source cell (red outline). The resulting spatial distribution shows the likelihood of transitions after three seconds, with darker blue indicating higher probability. Every cell in the discretized grid is equipped with this vehicle intention distribution.

Fig. 6 further shows that the learned vehicle intention distribution is broadly dispersed rather than dominated by a single next-step transition. Most cells exhibit high entropy (median 4.10 bits, IQR 3.75–4.26) and low dominant transition probability (median 0.171, IQR 0.143–0.243), with a strong negative correlation between the two quantities ($r = -0.943$).

As part of the offline stage, we also replay Algorithm 2 using telematics trajectories and official crash reports from the summer dataset (August 2024) to calibrate the alert threshold. Here the official crash reports are treated as proxy labels for safety-relevant abnormal events during threshold selection. Specifically, we sweep T across the range of

Table 1

Threshold vs. Precision, Recall and F1-Score.

Threshold	Precision	Recall	F1-score
0.000	0.186	1.000	0.313
16.084	0.619	1.000	0.765
18.076	0.600	0.923	0.727
20.442	0.632	0.923	0.750
23.142	0.667	0.923	0.774
26.410	0.706	0.923	0.800
28.182	0.750	0.923	0.828
30.000	0.800	0.923	0.857
39.008	0.750	0.692	0.720
60.000	0.727	0.615	0.667
62.300	0.889	0.615	0.727
90.000	0.875	0.538	0.667
120.000	1.000	0.538	0.700
150.000	1.000	0.462	0.632

observed accumulated risk scores and, for each candidate, compute precision, recall, and F1-score. Table 1 summarizes these results, showing that the maximum F1-score of 0.857 occurs at $T = 30$, which is adopted as the final alert threshold T^* .

3.3. Online risk score dynamics

In the online stage, each incoming telematics record $S_{i,t}^* = (\text{lat}_{i,t}, \text{lon}_{i,t}, v_{i,t})$ is immediately converted into its discretized state $S_{i,t} =$

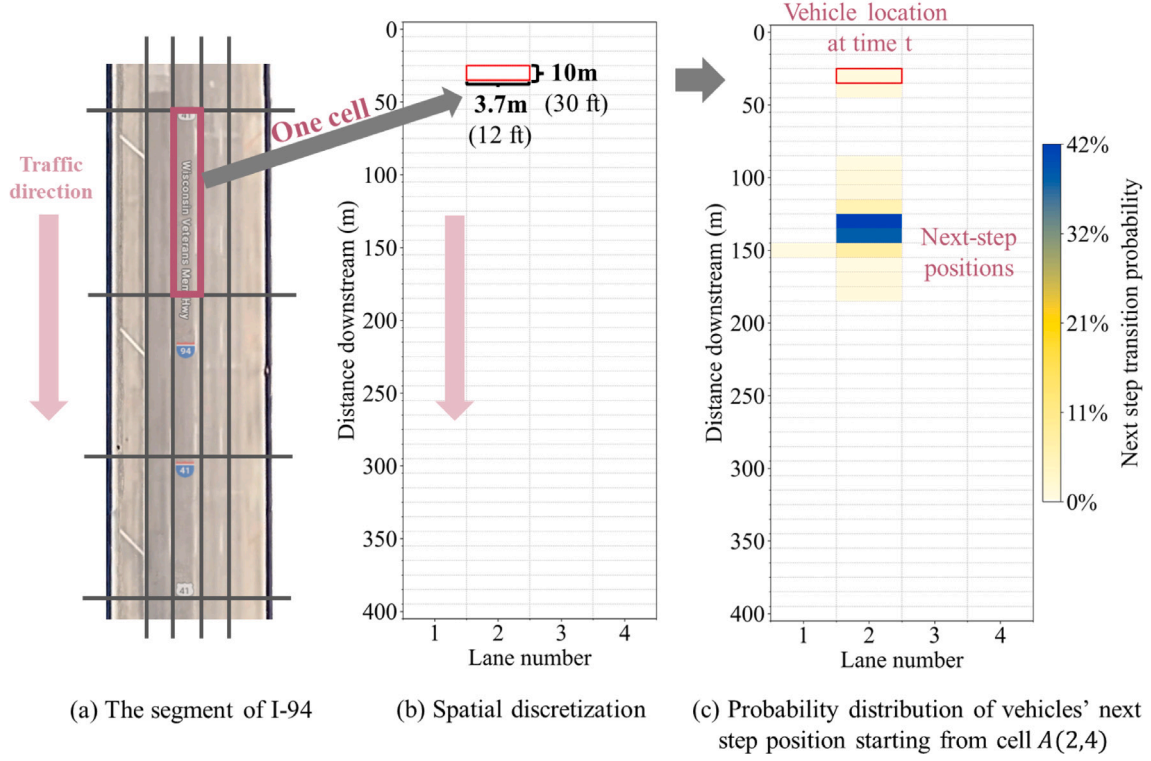


Fig. 5. Illustration of spatial-temporal discretization and vehicle intention distribution extraction.

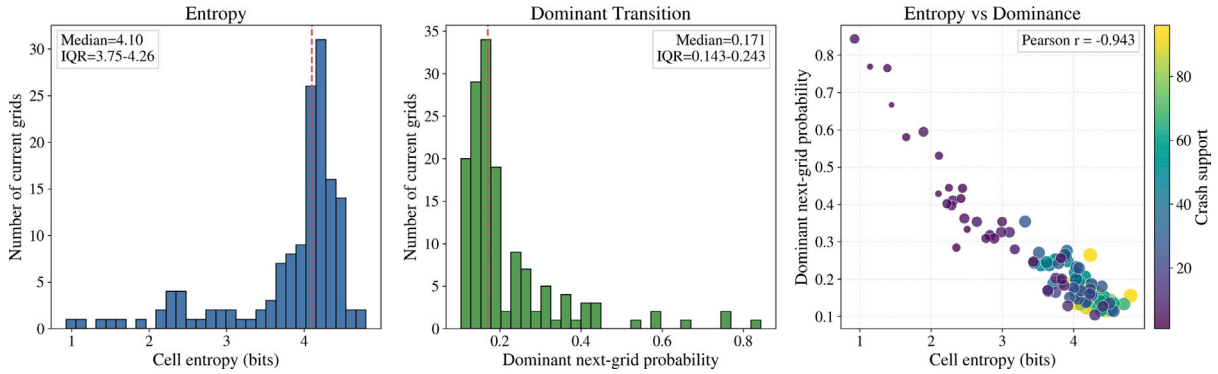


Fig. 6. Statistical characteristics of the extracted vehicle intention distribution across discretized cells.

$(x_{i,t}, y_{i,t}, v_{i,t})$ using the lane-matching algorithm (Algorithm 1) together with longitudinal segmentation. Once the spatial cell $(x_{i,t}, y_{i,t})$ is obtained, three complementary risk components $R_{i,t}^p$, $R_{i,t}^s$, and $R_{i,t}^\ell$ are computed at that cell. Because these components contribute differently to crash inference, we weight them with factors $w_p = 3$, $w_s = 2$, and $w_\ell = 4$, respectively, reflecting the relative importance of abrupt lateral maneuvers (highest weight), unusual position transitions, and abnormal speed deviation.

The instantaneous risk contributed by vehicle i at time t is therefore

$$R_{i,t}(x_{i,t}, y_{i,t}) = w_p R_{i,t}^p(x_{i,t}, y_{i,t}) + w_s R_{i,t}^s(x_{i,t}, y_{i,t}) + w_\ell R_{i,t}^\ell(x_{i,t}, y_{i,t}). \quad (8)$$

These per-vehicle risks are aggregated across all vehicles occupying the same cell to obtain the cell-level risk $\bar{R}_t(x, y)$, which is then accumulated through time via the risk map $A_t(x, y)$. If a vehicle traverses cell (x, y) at time t , the reset indicator $\phi_t(x, y) = 1$ clears the accumulated risk, indicating that the local hazard has been resolved. Whenever any cell satisfies $A_t(x, y) \geq T^*$, the system immediately triggers a lane-level abnormal traffic alert.

Fig. 7 illustrates the evolution of the lane-level risk map for the crash event discussed in the online stage. The first frame in this figure (06:36:08) shows normal traffic conditions with uniformly low risk levels. In the sequence of four frames of Fig. 7 (06:45:29 to 06:45:53), the system begins to capture the disturbance once new telematics records become available. Because only a very small number of vehicles passed the crash location between 06:43 and 06:45 due to data sparsity, the system could not observe any abnormal behavior during that window. When the density of observations increases at 06:45:29, likely due to the onset of congestion, the first visible signs of abnormal behavior appear, including simultaneous slowdowns and lane changes that are inconsistent with the offline intention distribution. These anomalies cause the risk to rise quickly, and by 06:45:41 the maximum score reaches $30.6 > T^* = 30$, which triggers an alert approximately 1.2 min before the official crash notification time. The final frame of Fig. 7 (06:47:35) shows continued growth of the risk around the crash location, matching the evolution of the actual crash and refining the detected lane-level location over time.

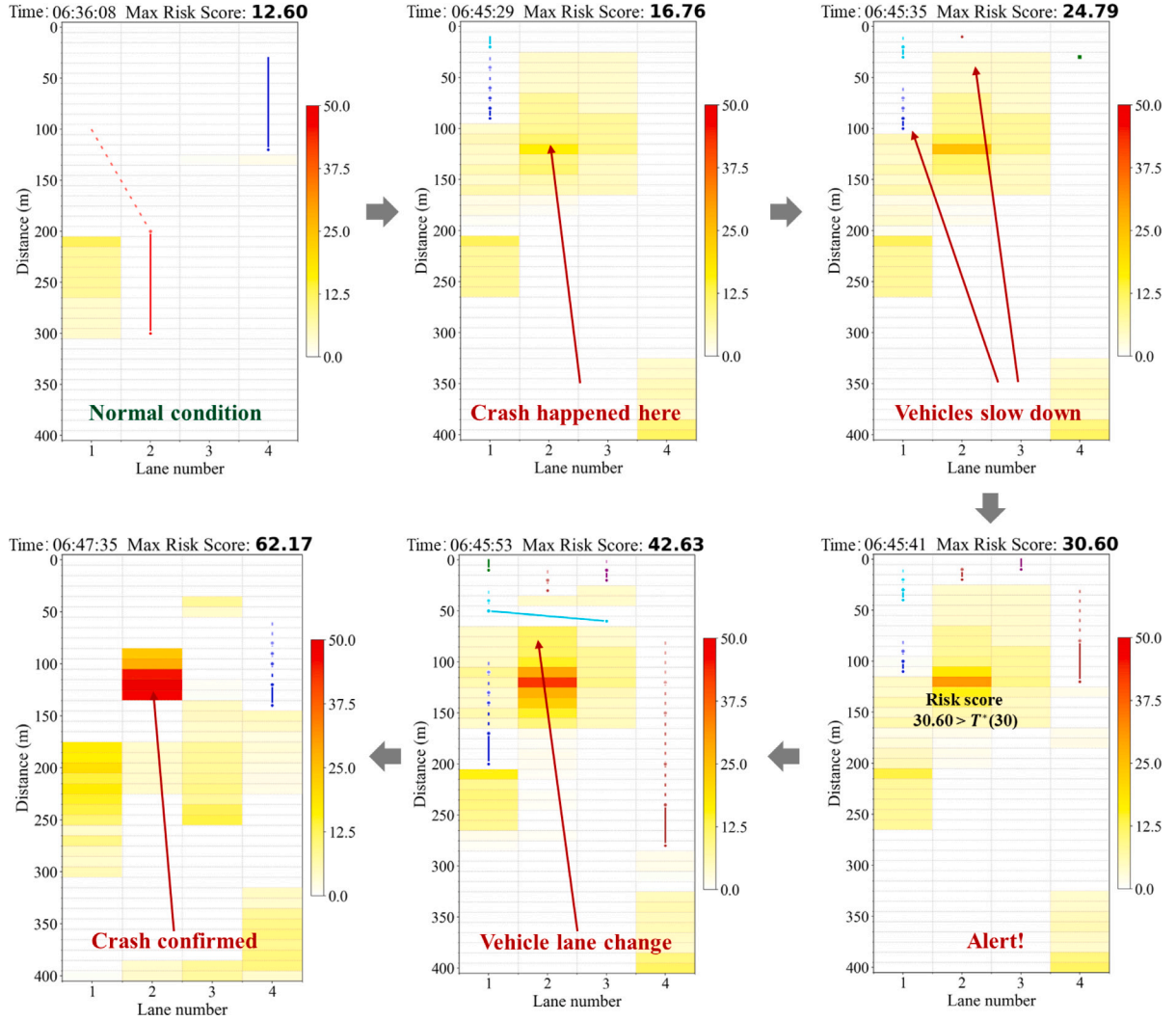


Fig. 7. Sequential lane-level risk map figures with annotated maximum risk scores.

3.4. Computational efficiency

The proposed online detection module is highly efficient. Even on a 2022 MacBook Air equipped with an Apple M2 chip and 8 GB of memory, each complete evaluation window is processed in well under one second. This real-time performance is enabled by the cell-based trajectory discretization and closed-form scoring modules, which eliminate iterative computation and maintain rapid response even when telematics observations are sparse.

3.5. Overall performance

To further evaluate the system performance, we extend the online validation to include both the summer and winter datasets. The alert threshold T^* and component weights (w_p, w_s, w_l), calibrated from the summer data, are applied directly to the winter dataset without retraining or parameter adjustment. This cross-temporal setting provides a temporally independent test of the calibrated framework. For empirical validation, official crash reports are used as proxy labels for abnormal events and as the reference source for event time and report time. As shown in Fig. 8(a), each case involves three timestamps: the recorded crash occurrence time t_{crash} , the crash report time t_{report} , and the model-generated alert time t_{alert} . Both delays are measured relative to the same crash-time reference:

$$\Delta t_{\text{report}} = t_{\text{report}} - t_{\text{crash}}, \quad \Delta t_{\text{detect}} = t_{\text{alert}} - t_{\text{crash}}.$$

Therefore, the reported lead time is not defined relative to the reporting time alone. Instead, Fig. 8(a) directly compares report delay and detection delay on a common event-time baseline. Many cases exhibit smaller detection delays than report delays, indicating that the system can identify emerging disturbances earlier than the official reporting process when sufficient telematics observations are available.

For comparison, non-crash cases are drawn from the remaining days within the same seven-day period that contain no recorded crashes. Intervals with missing telematics coverage, for example periods after 6:00 PM on January 15 when no telematics observations are available, are excluded to ensure consistent evaluation. Table 2 summarizes the combined detection results across all evaluated cases. Out of 83 report-labeled crash events used as proxy abnormal-event labels, 62 trigger alerts, yielding a proxy-event identification rate of 74.7%. Among 491 non-crash cases, only three false alarms occur, resulting in a precision of 95.4% and an overall F1 score of 0.84. The results are also shown in Fig. 8(b).

It is worth noting that the missed report-labeled cases mainly correspond to minor or operationally heterogeneous events that produce no observable disturbance in lane-level trajectories, such as vehicles pulling over or mechanical failures, or low-speed fender-benders occurring on the shoulder. Some report entries are safety-relevant but do not generate a measurable mainline traffic anomaly. For example, one undetected case in the official reports describes a vehicle pulling onto the right shoulder before catching fire, with no effect on surrounding

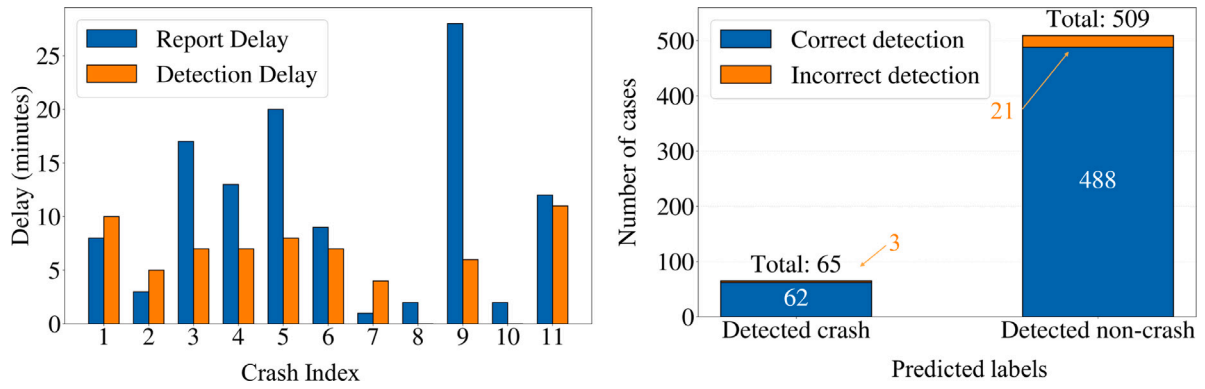


Fig. 8. Detection performance evaluation.

Table 2

Confusion matrix using official crash reports as proxy labels for abnormal-event evaluation.

Model output	Proxy label		Total
	Crash-report event	No crash-report event	
Alert	62	3	65
No alert	21	488	509
Total	83	491	574

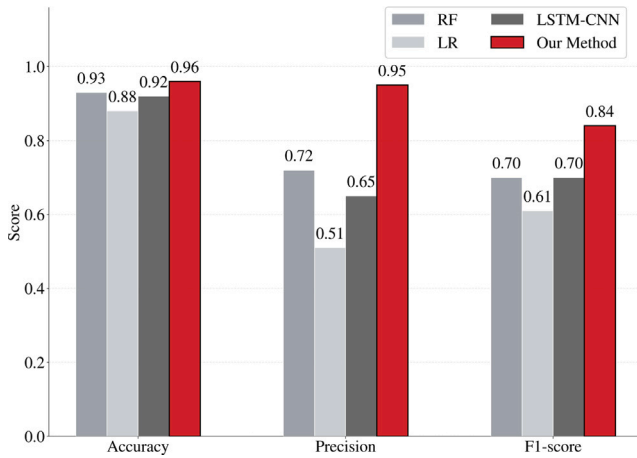


Fig. 9. Performance comparison with representative baseline methods.

traffic flow (“UNIT 1... BEGAN TO SMOKE AND CATCH FIRE AS UNIT 1 PULLED OVER TO THE RIGHT SHOULDER...”). Because this incident generated no measurable anomaly in lane-level positions, speeds, or transitions, it was not identifiable from sparse telematics trajectories. Consequently, the proposed framework is most effective for abnormal events that induce detectable behavior changes in the traffic stream.

To further assess the plausibility of lane-level localization, we compare the detected lane with the lane-related information recorded in crash reports for roadway segments where such information is available. In a random sample of 20 roadway segments, the detected lane agrees with the reported lane in 17 cases (85%). Among the remaining three cases, two differ from the reported lane by only one adjacent lane. Despite the substantial positional uncertainty in sparse telematics data, especially under high-speed freeway conditions, this agreement analysis still shows a high level of consistency and therefore provides practical support for the plausibility of lane-level localization in the proposed framework.

To further evaluate the relative performance of the proposed framework, we compare it with three representative baseline methods, including a Random Forest (RF) classifier (Roy and Hasan, 2026), a Logistic Regression (LR) classifier (Moriano et al., 2024), and an LSTM-CNN model (Li et al., 2020b). These baselines are selected to represent both

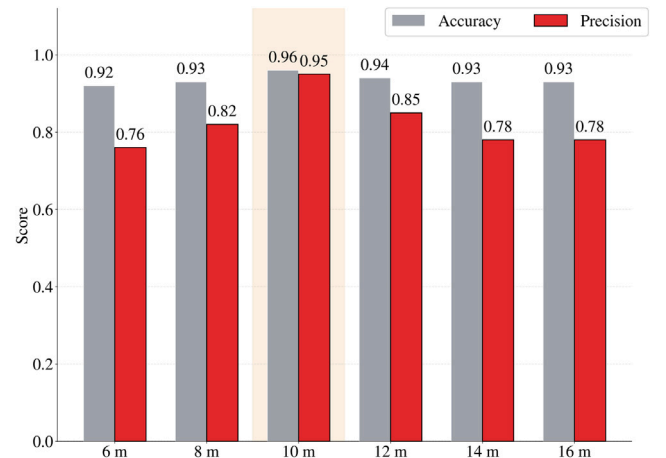


Fig. 10. Sensitivity analysis of detection performance under different longitudinal cell lengths.

classical machine-learning and deep spatiotemporal sequence-learning approaches commonly used in abnormal traffic detection research.

The comparison results are shown in Fig. 9. The proposed method achieves the best overall performance among all compared methods, with particularly clear advantages in precision and F1-score. These results show that the proposed framework outperforms the representative baseline approaches on the evaluated dataset.

4. Sensitivity analysis

4.1. Cell length

To further evaluate the robustness of the proposed framework with respect to spatial discretization, we conduct a sensitivity analysis on the longitudinal cell length. In the main experiments, the cell length is set to 10 m based on the discretization rationale described in Section 2.2.1. Here, we vary the cell length from 6 m to 16 m and re-evaluate the detection performance under each setting. The comparison results are shown in Fig. 10.

As illustrated in Fig. 10, the 10 m setting achieves the best overall performance among the tested configurations, with the highest

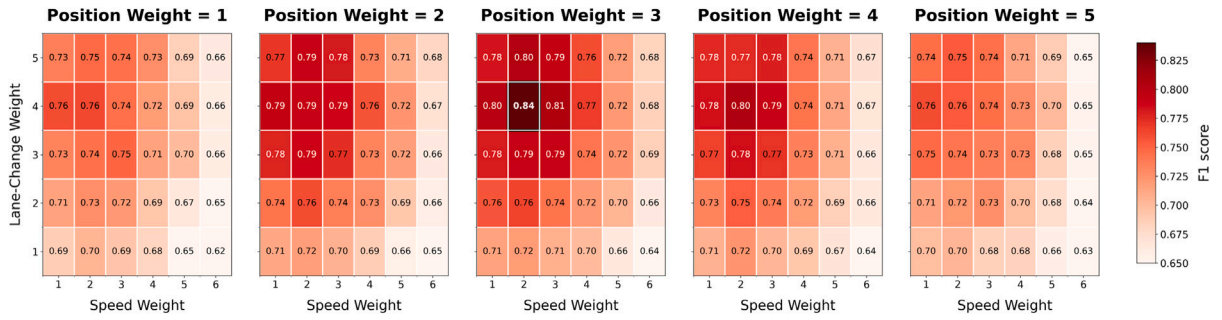


Fig. 11. Sensitivity analysis of the weighting coefficients in the total risk formulation.

accuracy and precision. Both shorter and longer cell lengths lead to some degree of performance degradation, indicating that the proposed framework is sensitive to the spatial discretization choice. Accordingly, 10 m is retained as the default cell length in the main experiments.

4.2. Weight sensitivity analysis

The total cell-level risk in Eq. (8) is defined as a weighted combination of the three complementary modules: position transition risk, speed deviation risk, and lateral maneuver risk. In the main experiments, we adopt $(w_p, w_s, w_\ell) = (3, 2, 4)$ and evaluate its robustness through sensitivity analysis. The linear form is used here as a parsimonious and interpretable first-order fusion rule under sparse telematics observations, rather than as a claim that the true interaction among the three risk components is strictly linear or statistically independent.

To further evaluate the robustness of this weighting design, we conduct a sensitivity analysis by varying the three coefficients over a reasonable range and re-evaluating the corresponding detection performance. Fig. 11 shows the resulting F1 scores for different combinations of (w_p, w_s, w_ℓ) . The best performance is achieved at (3, 2, 4), which is therefore retained as the default setting in this study. The performance also remains comparatively stable in the neighborhood of this configuration.

4.3. Peak-hour versus off-peak-hour analysis

To further examine the temporal heterogeneity of the proposed framework, we compare its detection performance under peak-hour and off-peak-hour conditions. In this study, peak-hour cases are defined as those whose reference timestamps fall within the two recurring freeway commuter peak periods, i.e., 7:00–9:00 and 16:00–18:00, while off-peak cases include the remaining hours.

Under this partition, the peak-hour subset contains 46 crash cases and 272 non-crash cases, for a total of 318 evaluated cases. The off-peak subset contains 37 crash cases and 219 non-crash cases, for a total of 256 evaluated cases. The comparison results are shown in Fig. 12.

As illustrated in Fig. 12, the proposed framework achieves higher recall and F1 score during peak hours, while off-peak hours yield slightly higher precision. Specifically, the peak-hour subset achieves an accuracy of 0.972, a precision of 0.951, a recall of 0.848, and an F1 score of 0.897. By comparison, the off-peak subset achieves an accuracy of 0.941, a precision of 0.958, a recall of 0.622, and an F1 score of 0.754. These results show a clear temporal difference in the recall–precision trade-off between peak-hour and off-peak-hour conditions.

5. Discussion

The experimental and supplementary analyses above reveal several consistent patterns that help interpret the proposed framework. First, the entropy and dominant-transition statistics indicate that normal freeway behavior in the discretized grid is inherently probabilistic and

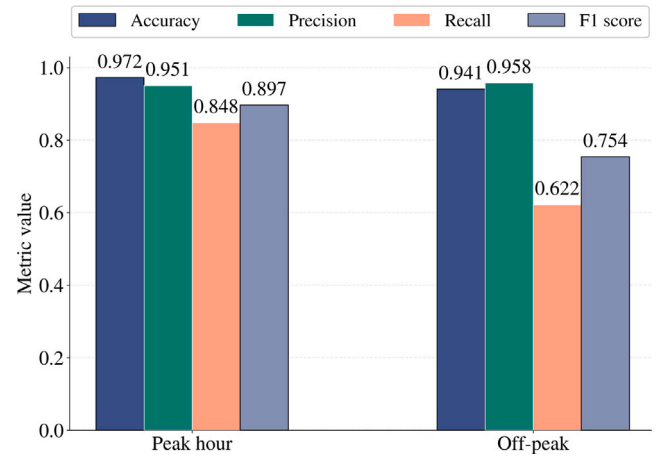


Fig. 12. Comparison of abnormal traffic detection performance under peak-hour and off-peak-hour conditions.

multi-modal rather than nearly deterministic. This supports the use of a distribution-based behavioral baseline, because a single deterministic next-state assumption would not adequately represent the observed variability in normal driving trajectories.

Second, the cell-length sensitivity analysis suggests that spatial discretization must balance local resolution against robustness under sparse telematics sampling. Very short cells tend to produce a more fragmented representation that is more sensitive to missing observations, whereas very long cells smooth out localized disturbances. The empirical optimum at 10 m therefore appears consistent with the freeway setting considered in this study, where vehicles move quickly but lane-level disturbances remain spatially localized.

Third, the weight-sensitivity analysis supports both the empirical stability and the behavioral interpretation of the selected weighting scheme. The neighborhood stability around $(w_p, w_s, w_\ell) = (3, 2, 4)$ indicates that the fusion framework is not excessively sensitive to moderate coefficient perturbations. At the same time, the ranking of the weights is consistent with the functional roles of the three modules: lateral maneuver risk captures immediate evasive behavior near a disturbance, position transition risk captures deviations from learned movement patterns, and speed deviation risk provides a broader but less specific signal that may also arise in recurrent congestion. Importantly, these modules are not assumed to be statistically independent. Some overlap is expected because abnormal lane changes, transition irregularities, and local speed perturbations often co-occur around the same disturbance. In this study, the weighted fusion should therefore be interpreted as an aggregation of correlated but behaviorally distinct evidence streams, adopted to improve robustness under sparse observations rather than to eliminate all possible double counting.

To further examine the aggregation mechanism beyond the weight heatmap, we additionally sweep the alert threshold while fixing the

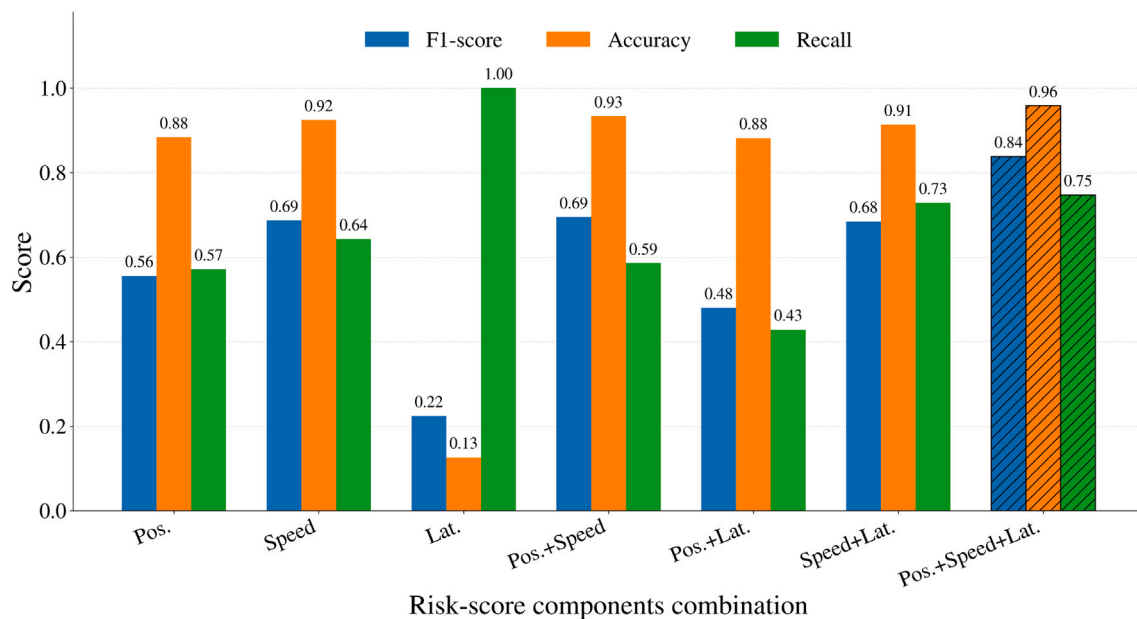


Fig. 13. Comparison of risk-score components under adaptive thresholds.

selected weight configuration. Fig. 13 shows that the full three-module fusion remains competitive across the main threshold region instead of depending on a single isolated operating point. This result does not rule out nonlinear coupling or threshold effects among the three components, but it suggests that the practical advantage of the current linear fusion is reasonably stable under threshold variation.

Finally, the peak-hour versus off-peak-hour comparison suggests that temporal traffic context affects the detectability of abnormal events. The higher recall and F1 score during peak hours may reflect stronger collective behavioral responses to lane-blocking disturbances, whereas the slightly higher off-peak precision may indicate a more conservative detection pattern when surrounding traffic is lighter. Together, these findings help clarify both the operating mechanism and the scope of applicability of the proposed framework.

6. Conclusion

This study introduces a real-time abnormal traffic detection framework that relies exclusively on sparse telematics data and road geometry from OSM. In the offline stage, we perform low-cost lane matching using a vector cross-product method and discretize vehicle trajectories into spatial cells to mitigate data sparsity. During the online stage, each incoming telematics record is evaluated for position transition anomalies, speed deviations, and abnormal lane-change maneuvers, and these risk components are accumulated in a lane-specific map. The framework is designed primarily to enable robust online detection under sparse observations, while lane-level localization is provided as an important output capability of the resulting risk map. When the accumulated risk in any cell surpasses the threshold determined offline via F1-score optimization, the system issues an immediate alert with precise lane-level localization, enabling timely warnings that can help prevent secondary crashes, expedite response, and reduce associated economic losses and casualties.

To comprehensively validate the framework, we conducted experiments on combined summer (August 2024) and winter (January 2024) datasets covering freeway segments across Wisconsin, using official crash reports as proxy labels and evaluation benchmarks for abnormal events. A total of 83 crash events and 491 non-crash control periods were analyzed. The proposed method successfully generated alerts for 62 report-labeled events while producing only three false alarms, yielding a detection rate of 74.7%, a precision of 95.4%, an overall F1-score of 0.84, and an accuracy of 96.0%. Notably, 13% of the detected

crashes were identified 3 min before the official notification time. The missed proxy-labeled cases were primarily minor or shoulder-related events that caused no measurable disturbance in lane-level trajectories, whereas all major lane-blocking cases were accurately localized to the correct lanes. These results demonstrate that the framework generalizes well across seasons and maintains robust detection and localization performance under diverse traffic and environmental conditions.

Future work will explore learning-based intention models that integrate kinematic structure and historical trajectory features. At present, parameters such as risk module weights are determined through empirical tuning, and learning-based methods could automatically optimize these quantities from data. Future work may also investigate redundancy-aware or nonlinear fusion mechanisms, so that interaction terms, threshold effects, or learned aggregation rules can be estimated more formally once larger labeled datasets become available. As the dataset grows, threshold calibration may also be refined to account for peak and off-peak periods, roadway types, and regional driving patterns. With greater computational resources, the framework could further adapt its parameters in real time to improve detection speed and accuracy.

CRedit authorship contribution statement

Shixiao Liang: Writing – original draft, Visualization, Validation, Investigation. **Chengyuan Ma:** Writing – review & editing, Conceptualization. **Pei Li:** Writing – review & editing, Data curation. **Haotian Shi:** Resources, Data curation. **Jiaxi Liu:** Investigation, Data curation. **Hang Zhou:** Writing – review & editing, Data curation. **Keke Long:** Writing – review & editing, Validation. **Bofeng Cao:** Investigation. **Todd Szymkowski:** Supervision, Data curation. **Xiaowei Shi:** Formal analysis, Data curation. **Xiaopeng Li:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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